

11. Let $b = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$, and let A be the matrix in Exercise 9. Is b in the range of the linear transformation $x \mapsto Ax$? Why or why not?

Solution: We need to find out if there exists an x s.t. $Ax = b$.

By Theorem 3, the matrix equation $Ax = b$ has the same solution as the system of equations corresponding to the augmented matrix $[A \ b]$.

$$\begin{bmatrix} -1 \\ 0 \end{bmatrix} \xrightarrow{\text{Row 2} = \text{Row 2} - 2 \times \text{Row 1}} \begin{bmatrix} -1 \\ 2 \end{bmatrix} \xrightarrow{\text{Row 2} = \text{Row 2} - 2 \times \text{Row 1}} \begin{bmatrix} -1 \\ 0 \end{bmatrix} \xrightarrow{\text{Row 1} = \text{Row 1} + 4 \times \text{Row 2}} \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

The final augmented matrix will be:

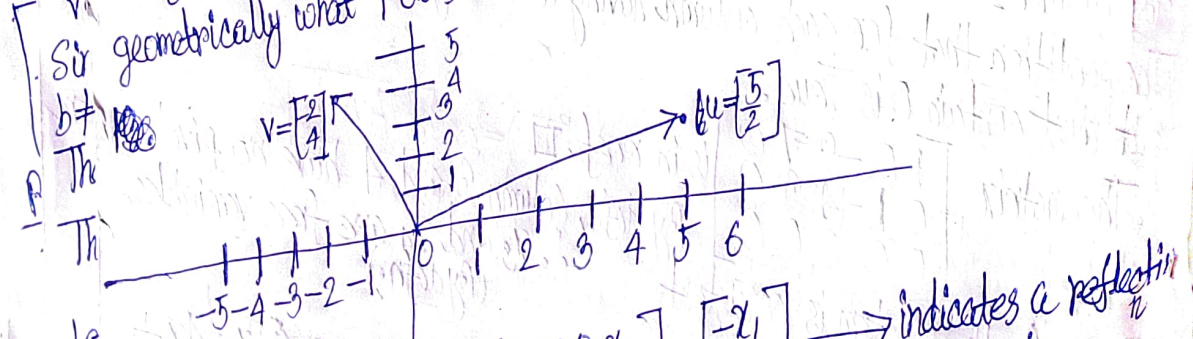
$$\begin{bmatrix} 1 & 0 & -9 & 7 & 3 \\ 0 & 1 & -4 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

From Prob 9, we can easily prove that the final augmented matrix is in ref. There are no rows in the matrix of the form $[0 \ 0 \ 0 \ \dots \ 0 \ b]$ where $b \neq 0$.

By Theorem 2, the system of equations is consistent, i.e. a solution exists.

$\therefore b$ is in the range of the linear transformation. $\square$

In Exercises 13-16, use a rectangular coordinate system to plot $u = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$, $v = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$, and their images under the given transformation T . (Make a separate and reasonably large sketch for each exercise). Describe geometrically what T does to each vector x in $\mathbb{R}^2$.



13) $T(x) = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 \cdot x_1 + 0 \cdot x_2 \\ 0 \cdot x_1 + (-1) \cdot x_2 \end{bmatrix} = \begin{bmatrix} -x_1 \\ -x_2 \end{bmatrix}$ — indicates a reflection about the origin.

$$15) T(x) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \cdot x_1 + 0 \cdot x_2 \\ 0 \cdot x_1 + 1 \cdot x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ x_2 \end{bmatrix}$$

This indicates a projection onto the x_2 axis.